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# RICHMACK

## Game Theory Strategies

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A Practical Field Guide to Cooperation, Defection, Reciprocity,  
Learning, and Evolutionary Competition

Companion volume to JinnLab  
Axelrod Python 4.14.0 Edition

Jeremy Franklin (Richmack)

2026



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# Publisher's Note

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This A5 pocket edition is designed as a compact field manual: theory first, then strategy families, historical tournament entries, experimental methods, and a reference catalog. Gold-framed cards identify strategy profiles; shaded callouts identify interpretation rules and experimental cautions.<sup>1</sup>

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<sup>1</sup>The strategy catalog and naming conventions in this edition follow the Axelrod Python 4.14.0 documentation and the JinnLab strategy catalog built around that release.

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# About This Edition

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This book is an original explanatory companion to JinnLab. It organizes and interprets the strategy ecosystem made available by the Axelrod Python library rather than reproducing the library documentation. The Axelrod 4.14.0 documentation reports 243 available strategies and describes the project as a reproducible framework for Iterated Prisoner's Dilemma research.

Strategy behavior can depend on parameters, match length, noise, random seed, opponent composition, payoff rules, and library version. Treat every ranking in JinnLab as evidence about a particular experiment, not a universal hierarchy.

Copyright © 2026 Jeremy Franklin. Explanatory text and JinnLab exercises are original. Strategy names, historical attributions, and algorithmic ideas belong to their respective authors and research traditions. Axelrod Python is an open-source project distributed under the MIT License.

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# Preface: Strategy Is Conditional

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Game theory becomes useful when we stop asking, “What is the best strategy?” and start asking, “Best against whom, for how long, under what information, with what error rate, and inside what population?” The Iterated Prisoner’s Dilemma is a compact laboratory for those questions.

JinnLab turns the catalog into an experimental environment. The book supplies the conceptual layer: what each family is trying to accomplish, which tradeoffs it makes, and what evidence to collect before drawing conclusions.

## How to Read a Strategy Profile

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## The Standard Prisoner’s Dilemma

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The exact numbers can vary, but the ordering is the point: unilateral defection is tempting, mutual cooperation beats mutual defection, and the social optimum conflicts with the immediate individual incentive.

## Experimental Discipline

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- Use multiple seeds for stochastic strategies. A single random realization is an anecdote.
- Increase repetitions before treating a small payoff difference as meaningful.



- Change the opponent field. Tournament ranking is a property of both the strategy and the population it faces.
- Test noise. Strategies that look excellent in perfect information can fail when one accidental action starts a retaliation loop.
- Use evolution separately from tournaments. High pairwise payoff and long-run population survival are related but not identical questions.

# Cooperation and the Logic of Repetition

|       | Opponent C | Opponent D |
|-------|------------|------------|
| You C | 3, 3       | 0, 5       |
| You D | 5, 0       | 1, 1       |

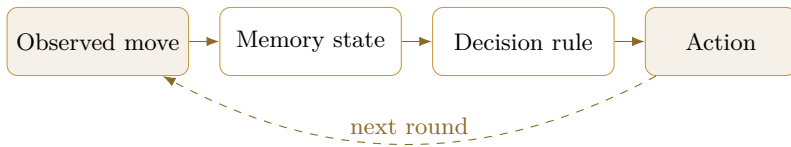
**Figure 1.1:** Standard Prisoner's Dilemma payoff ordering used throughout the book. The ordering matters more than the exact numbers.

In a one-shot dilemma, defection is individually attractive. Repetition changes the geometry of the problem because present behavior can affect future treatment. Reputation, retaliation, forgiveness, and credible punishment become strategic resources.

Long horizons favor strategies that can create stable cooperation without becoming easy targets. Short horizons increase the value of exploitation and make end-game behavior more important.

# Memory, Noise, and Information

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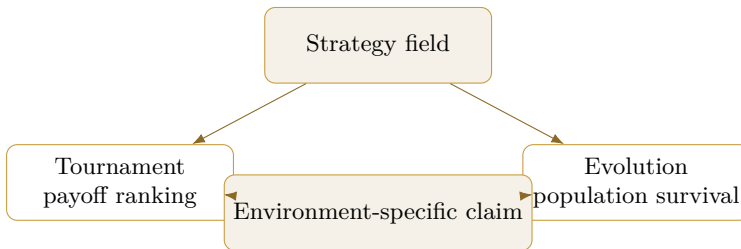
**Figure 2.1:** A compact model of repeated-game decision making: observations update state, state informs a rule, and the rule produces the next action. Noise can enter at observation or execution.

Memory depth determines how much of the interaction a strategy can condition on. More memory permits richer distinctions, but complexity does not automatically create better performance.

Noise creates a second problem: an observed defection may not express intent. Robust strategies therefore need error correction, forgiveness, or a way to distinguish persistent hostility from isolated mistakes.

# Tournaments, Populations, and Evolution

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**Figure 3.1:** Tournament success and evolutionary persistence answer different questions. JinnLab treats both as evidence about a specified environment, not universal proof of superiority.

A tournament asks which strategies collect the most payoff in a specified opponent field. A population process asks which behaviors reproduce or persist as the population changes. These are different experiments.

The Axelrod project was created to make such studies reproducible, including head-to-head matches, tournaments, and population dynamics.<sup>1</sup>

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<sup>1</sup>For the modern reproducible software framework, see Knight et al., 'An Open Reproducible Framework for the Study of the Iterated Prisoner's Dilemma' (2016).

# The Historical Axelrod Tournaments

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Robert Axelrod’s computer tournaments made direct reciprocity famous because Tit for Tat performed remarkably well despite its simplicity. The first-tournament implementations in Axelrod Python are reconstructed from published descriptions, while the second-tournament family preserves a larger historical strategy set.<sup>1</sup>

Recent reproduction work on the second tournament reinforces a central lesson: rankings are robust in some settings, but the composition of the field materially shapes what “wins.”<sup>2</sup>

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<sup>1</sup>Robert Axelrod’s tournament work established the iterated Prisoner’s Dilemma as a canonical setting for studying direct reciprocity; see Axelrod (1980) and – for the broader synthesis – *The Evolution of Cooperation* (1984).

<sup>2</sup>See Knight, Campbell, Harper, Gaffney, and Glynatsi, ‘Reviving, Reproducing, and Revisiting Axelrod’s Second Tournament’ (2025).

# Reciprocity: Tit-for-Tat and Its Descendants

|          |   |   |   |   |   |   |
|----------|---|---|---|---|---|---|
|          | 1 | 2 | 3 | 4 | 5 | 6 |
| Opponent | C | C | D | C | C | D |
| TFT      | C | C | C | D | C | C |

**Figure 5.1:** Tit for Tat in one line: cooperate first, then copy the opponent’s previous move. The one-round lag creates both retaliation and forgiveness.

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

## Tit For Tat

**CORE IDEA**

Cooperate first, then mirror the opponent’s previous action.

**STRATEGIC CHARACTER**

Reciprocity

**JINNLAB EXPERIMENT**

Test it against Defector, Cooperator, Grudger, and a noisy opponent. Watch how quickly retaliation begins and how quickly cooperation recovers.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Tit For 2 Tats

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Tit For 2 Tats against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Two Tits For Tat

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Two Tits For Tat against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Hard Tit For Tat

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Hard Tit For Tat against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Hard Tit For 2 Tats

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Hard Tit For 2 Tats against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Suspicious Tit For Tat

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Suspicious Tit For Tat against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Anti Tit For Tat

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Anti Tit For Tat against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Bully

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Bully against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Contrite Tit For Tat

### CORE IDEA

A reciprocity strategy designed to repair cooperation after accidental defections.

### STRATEGIC CHARACTER

Noise-aware reciprocity

### JINNLAB EXPERIMENT

Compare directly with Tit For Tat at non-zero noise and inspect cooperation rate.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Adaptive Tit For Tat

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Adaptive Tit For Tat against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Random Tit For Tat

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Random Tit For Tat against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Sneaky Tit For Tat

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Sneaky Tit For Tat against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Spiteful Tit For Tat

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Spiteful Tit For Tat against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Dynamic Two Tits For Tat

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Dynamic Two Tits For Tat against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Slow Tit For Two Tats 2

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Slow Tit For Two Tats 2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Omega TFT

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Omega TFT against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Gradual

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Gradual against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Original Gradual

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Original Gradual against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Alexei

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Alexei against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Michaelos

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Michaelos against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Eugene Nier

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Eugene Nier against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Burn Both Ends

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run Burn Both Ends against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## N Tits For M Tats

### CORE IDEA

Strategy in the Tit-for-Tat family, using imitation, retaliation, forgiveness, or variations on direct reciprocity.

### STRATEGIC CHARACTER

Reciprocity

### JINNLAB EXPERIMENT

Run N Tits For M Tats against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Cooperation, Defection, and Simple Baselines

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Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

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### Cooperator

#### CORE IDEA

Always cooperate.

#### STRATEGIC CHARACTER

Unconditional cooperation

#### JINNLAB EXPERIMENT

Use it as the generosity baseline. Compare its payoff against Defector and against itself.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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### Defector

#### CORE IDEA

Always defect.

**STRATEGIC CHARACTER**

Unconditional defection

**JINNLAB EXPERIMENT**

Use it as the exploitation baseline. Compare short and long matches against forgiving strategies.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Random****CORE IDEA**

Choose actions probabilistically rather than from a deterministic rule.

**STRATEGIC CHARACTER**

Stochastic baseline

**JINNLAB EXPERIMENT**

Run many seeds. A single match is not representative; compare distributions across repetitions.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Alternator****CORE IDEA**

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

**STRATEGIC CHARACTER**

Cooperation, Defection, and Simple Baselines

### JINNLAB EXPERIMENT

Run Alternator against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

▮ *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Tricky Cooperator

### CORE IDEA

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

### STRATEGIC CHARACTER

Cooperation, Defection, and Simple Baselines

### JINNLAB EXPERIMENT

Run Tricky Cooperator against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

▮ *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Tricky Defector

### CORE IDEA

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

### STRATEGIC CHARACTER

Cooperation, Defection, and Simple Baselines

### JINNLAB EXPERIMENT

Run Tricky Defector against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Better and Better

### CORE IDEA

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

### STRATEGIC CHARACTER

Cooperation, Defection, and Simple Baselines

### JINNLAB EXPERIMENT

Run Better and Better against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Very Bad

### CORE IDEA

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

### STRATEGIC CHARACTER

Cooperation, Defection, and Simple Baselines

### JINNLAB EXPERIMENT

Run Very Bad against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

▮ *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Short Mem

### CORE IDEA

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

### STRATEGIC CHARACTER

Cooperation, Defection, and Simple Baselines

### JINNLAB EXPERIMENT

Run Short Mem against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

▮ *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Inverse

### CORE IDEA

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

### STRATEGIC CHARACTER

Cooperation, Defection, and Simple Baselines

**JINNLAB EXPERIMENT**

Run Inverse against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Negation

**CORE IDEA**

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

**STRATEGIC CHARACTER**

Cooperation, Defection, and Simple Baselines

**JINNLAB EXPERIMENT**

Run Negation against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Willing

**CORE IDEA**

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

**STRATEGIC CHARACTER**

Cooperation, Defection, and Simple Baselines

### JINNLAB EXPERIMENT

Run Willing against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

▮ *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Hopeless

### CORE IDEA

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

### STRATEGIC CHARACTER

Cooperation, Defection, and Simple Baselines

### JINNLAB EXPERIMENT

Run Hopeless against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

▮ *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Desperate

### CORE IDEA

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

### STRATEGIC CHARACTER

Cooperation, Defection, and Simple Baselines



**JINNLAB EXPERIMENT**

Run Desperate against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Appeaser****CORE IDEA**

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

**STRATEGIC CHARACTER**

Cooperation, Defection, and Simple Baselines

**JINNLAB EXPERIMENT**

Run Appeaser against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Doubler****CORE IDEA**

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

**STRATEGIC CHARACTER**

Cooperation, Defection, and Simple Baselines

## JINNLAB EXPERIMENT

Run Doubler against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Momentum

### CORE IDEA

Simple behavioral baseline useful for understanding what more complex strategies are actually accomplishing.

### STRATEGIC CHARACTER

Cooperation, Defection, and Simple Baselines

## JINNLAB EXPERIMENT

Run Momentum against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Grudges, Punishment, and Retaliation

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Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

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## Grudger

### CORE IDEA

Cooperate until the opponent defects once, then punish indefinitely.

### STRATEGIC CHARACTER

Permanent retaliation

### JINNLAB EXPERIMENT

Introduce one defection or noise event and observe whether cooperation can ever return.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Forgetful Grudger

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Forgetful Grudger against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Soft Grudger

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Soft Grudger against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## General Soft Grudger

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run General Soft Grudger against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

## Grudger Alternator

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Grudger Alternator against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

## Opposite Grudger

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

**STRATEGIC CHARACTER**

Grudges, Punishment, and Retaliation

**JINNLAB EXPERIMENT**

Run Opposite Grudger against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Aggravater

**CORE IDEA**

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

**STRATEGIC CHARACTER**

Grudges, Punishment, and Retaliation

**JINNLAB EXPERIMENT**

Run Aggravater against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## CAPRI

**CORE IDEA**

A memory-three rule set built around cooperation, acceptance, punishment, recovery, and otherwise defecting.

### STRATEGIC CHARACTER

Error-correcting reciprocity

### JINNLAB EXPERIMENT

Run CAPRI against Tit For Tat, Defector, and itself; inspect whether punishment is limited and recovery succeeds.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Easy Go

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Easy Go against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Spiteful CC

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

**STRATEGIC CHARACTER**

Grudges, Punishment, and Retaliation

**JINNLAB EXPERIMENT**

Run Spiteful CC against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Gradual Killer

**CORE IDEA**

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

**STRATEGIC CHARACTER**

Grudges, Punishment, and Retaliation

**JINNLAB EXPERIMENT**

Run Gradual Killer against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Punisher

**CORE IDEA**



Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Punisher against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Inverse Punisher

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Inverse Punisher against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Level Punisher

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Level Punisher against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Tricky Level Punisher

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Tricky Level Punisher against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Retaliate

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Retaliate against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Retaliate 2

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Retaliate 2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Retaliate 3

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

**STRATEGIC CHARACTER**

Grudges, Punishment, and Retaliation

**JINNLAB EXPERIMENT**

Run Retaliate 3 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Limited Retaliate

**CORE IDEA**

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

**STRATEGIC CHARACTER**

Grudges, Punishment, and Retaliation

**JINNLAB EXPERIMENT**

Run Limited Retaliate against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Limited Retaliate 2

**CORE IDEA**

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Limited Retaliate 2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

## Limited Retaliate 3

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Limited Retaliate 3 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

## Fool Me Once

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Fool Me Once against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Forgetful Fool Me Once

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### STRATEGIC CHARACTER

Grudges, Punishment, and Retaliation

### JINNLAB EXPERIMENT

Run Forgetful Fool Me Once against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Once Bitten

### CORE IDEA

Strategy that emphasizes punishment, retaliation, or conditional forgiveness after defection.

### **STRATEGIC CHARACTER**

Grudges, Punishment, and Retaliation

### **JINNLAB EXPERIMENT**

Run Once Bitten against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Majority, Frequency, and Population Reading

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Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

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### Go By Majority

#### CORE IDEA

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

#### STRATEGIC CHARACTER

Majority, Frequency, and Population Reading

#### JINNLAB EXPERIMENT

Run Go By Majority against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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### Go By Majority 5



**CORE IDEA**

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

**STRATEGIC CHARACTER**

Majority, Frequency, and Population Reading

**JINNLAB EXPERIMENT**

Run Go By Majority 5 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Go By Majority 10

**CORE IDEA**

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

**STRATEGIC CHARACTER**

Majority, Frequency, and Population Reading

**JINNLAB EXPERIMENT**

Run Go By Majority 10 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Go By Majority 20

### CORE IDEA

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

### STRATEGIC CHARACTER

Majority, Frequency, and Population Reading

### JINNLAB EXPERIMENT

Run Go By Majority 20 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Go By Majority 40

### CORE IDEA

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

### STRATEGIC CHARACTER

Majority, Frequency, and Population Reading

### JINNLAB EXPERIMENT

Run Go By Majority 40 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Hard Go By Majority

**CORE IDEA**

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

**STRATEGIC CHARACTER**

Majority, Frequency, and Population Reading

**JINNLAB EXPERIMENT**

Run Hard Go By Majority against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Hard Go By Majority 5

**CORE IDEA**

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

**STRATEGIC CHARACTER**

Majority, Frequency, and Population Reading

**JINNLAB EXPERIMENT**

Run Hard Go By Majority 5 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Hard Go By Majority 10

### CORE IDEA

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

### STRATEGIC CHARACTER

Majority, Frequency, and Population Reading

### JINNLAB EXPERIMENT

Run Hard Go By Majority 10 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Hard Go By Majority 20

### CORE IDEA

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

### STRATEGIC CHARACTER

Majority, Frequency, and Population Reading

### JINNLAB EXPERIMENT

Run Hard Go By Majority 20 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Hard Go By Majority 40

**CORE IDEA**

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

**STRATEGIC CHARACTER**

Majority, Frequency, and Population Reading

**JINNLAB EXPERIMENT**

Run Hard Go By Majority 40 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Frequency Analyzer

**CORE IDEA**

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

**STRATEGIC CHARACTER**

Majority, Frequency, and Population Reading

**JINNLAB EXPERIMENT**

Run Frequency Analyzer against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Average Copier

### CORE IDEA

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

### STRATEGIC CHARACTER

Majority, Frequency, and Population Reading

### JINNLAB EXPERIMENT

Run Average Copier against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Nice Average Copier

### CORE IDEA

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

### STRATEGIC CHARACTER

Majority, Frequency, and Population Reading

### JINNLAB EXPERIMENT

Run Nice Average Copier against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## SelfSteem

**CORE IDEA**

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

**STRATEGIC CHARACTER**

Majority, Frequency, and Population Reading

**JINNLAB EXPERIMENT**

Run SelfSteem against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Grumpy

**CORE IDEA**

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

**STRATEGIC CHARACTER**

Majority, Frequency, and Population Reading

**JINNLAB EXPERIMENT**

Run Grumpy against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Stalker

### CORE IDEA

Strategy that summarizes the opponent's history statistically rather than reacting to only one move.

### STRATEGIC CHARACTER

Majority, Frequency, and Population Reading

### JINNLAB EXPERIMENT

Run Stalker against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Patterns, Cycles, and Mathematical Sequences

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Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

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### Cycler

#### CORE IDEA

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

#### STRATEGIC CHARACTER

Patterns, Cycles, and Mathematical Sequences

#### JINNLAB EXPERIMENT

Run Cycler against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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### Cycler DC

### CORE IDEA

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

### STRATEGIC CHARACTER

Patterns, Cycles, and Mathematical Sequences

### JINNLAB EXPERIMENT

Run Cycler DC against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Cycler DDC

### CORE IDEA

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

### STRATEGIC CHARACTER

Patterns, Cycles, and Mathematical Sequences

### JINNLAB EXPERIMENT

Run Cycler DDC against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Cycler CCD

**CORE IDEA**

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

**STRATEGIC CHARACTER**

Patterns, Cycles, and Mathematical Sequences

**JINNLAB EXPERIMENT**

Run Cycler CCD against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Cycler CCCD

**CORE IDEA**

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

**STRATEGIC CHARACTER**

Patterns, Cycles, and Mathematical Sequences

**JINNLAB EXPERIMENT**

Run Cycler CCCD against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Cycler CCCDCD

### CORE IDEA

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

### STRATEGIC CHARACTER

Patterns, Cycles, and Mathematical Sequences

### JINNLAB EXPERIMENT

Run Cyler CCCDCD against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Cyler CCCCCD

### CORE IDEA

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

### STRATEGIC CHARACTER

Patterns, Cycles, and Mathematical Sequences

### JINNLAB EXPERIMENT

Run Cyler CCCCCD against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## AntiCyler

**CORE IDEA**

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

**STRATEGIC CHARACTER**

Patterns, Cycles, and Mathematical Sequences

**JINNLAB EXPERIMENT**

Run AntiCycler against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Evolvable Cycler

**CORE IDEA**

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

**STRATEGIC CHARACTER**

Patterns, Cycles, and Mathematical Sequences

**JINNLAB EXPERIMENT**

Run Evolvable Cycler against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Sequence Player

### CORE IDEA

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

### STRATEGIC CHARACTER

Patterns, Cycles, and Mathematical Sequences

### JINNLAB EXPERIMENT

Run Sequence Player against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Thue-Morse

### CORE IDEA

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

### STRATEGIC CHARACTER

Patterns, Cycles, and Mathematical Sequences

### JINNLAB EXPERIMENT

Run Thue-Morse against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Thue-Morse Inverse

**CORE IDEA**

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

**STRATEGIC CHARACTER**

Patterns, Cycles, and Mathematical Sequences

**JINNLAB EXPERIMENT**

Run Thue-Morse Inverse against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Co to Do Ratio

**CORE IDEA**

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

**STRATEGIC CHARACTER**

Patterns, Cycles, and Mathematical Sequences

**JINNLAB EXPERIMENT**

Run Co to Do Ratio against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Golden

### CORE IDEA

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

### STRATEGIC CHARACTER

Patterns, Cycles, and Mathematical Sequences

### JINNLAB EXPERIMENT

Run Golden against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Pi

### CORE IDEA

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

### STRATEGIC CHARACTER

Patterns, Cycles, and Mathematical Sequences

### JINNLAB EXPERIMENT

Run Pi against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## e



### CORE IDEA

Strategy built around a repeatable sequence, cycle, or target ratio rather than direct reciprocity alone.

### STRATEGIC CHARACTER

Patterns, Cycles, and Mathematical Sequences

### JINNLAB EXPERIMENT

Run e against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Hunters, Probers, and Opponent Detection

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Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

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## Alternator Hunter

### CORE IDEA

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

### STRATEGIC CHARACTER

Hunters, Probers, and Opponent Detection

### JINNLAB EXPERIMENT

Run Alternator Hunter against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Cooperator Hunter

**CORE IDEA**

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

**STRATEGIC CHARACTER**

Hunters, Probers, and Opponent Detection

**JINNLAB EXPERIMENT**

Run Cooperator Hunter against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Cycle Hunter

**CORE IDEA**

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

**STRATEGIC CHARACTER**

Hunters, Probers, and Opponent Detection

**JINNLAB EXPERIMENT**

Run Cycle Hunter against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Defector Hunter

## CORE IDEA

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

## STRATEGIC CHARACTER

Hunters, Probers, and Opponent Detection

## JINNLAB EXPERIMENT

Run Defector Hunter against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Eventual Cycle Hunter

## CORE IDEA

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

## STRATEGIC CHARACTER

Hunters, Probers, and Opponent Detection

## JINNLAB EXPERIMENT

Run Eventual Cycle Hunter against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Math Constant Hunter

**CORE IDEA**

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

**STRATEGIC CHARACTER**

Hunters, Probers, and Opponent Detection

**JINNLAB EXPERIMENT**

Run Math Constant Hunter against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Random Hunter

**CORE IDEA**

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

**STRATEGIC CHARACTER**

Hunters, Probers, and Opponent Detection

**JINNLAB EXPERIMENT**

Run Random Hunter against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Prober

### CORE IDEA

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

### STRATEGIC CHARACTER

Hunters, Probers, and Opponent Detection

### JINNLAB EXPERIMENT

Run Prober against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Prober 2

### CORE IDEA

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

### STRATEGIC CHARACTER

Hunters, Probers, and Opponent Detection

### JINNLAB EXPERIMENT

Run Prober 2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Prober 3

**CORE IDEA**

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

**STRATEGIC CHARACTER**

Hunters, Probers, and Opponent Detection

**JINNLAB EXPERIMENT**

Run Prober 3 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Prober 4

**CORE IDEA**

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

**STRATEGIC CHARACTER**

Hunters, Probers, and Opponent Detection

**JINNLAB EXPERIMENT**

Run Prober 4 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Hard Prober

### CORE IDEA

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

### STRATEGIC CHARACTER

Hunters, Probers, and Opponent Detection

### JINNLAB EXPERIMENT

Run Hard Prober against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Naive Prober

### CORE IDEA

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

### STRATEGIC CHARACTER

Hunters, Probers, and Opponent Detection

### JINNLAB EXPERIMENT

Run Naive Prober against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Remorseful Prober



**CORE IDEA**

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

**STRATEGIC CHARACTER**

Hunters, Probers, and Opponent Detection

**JINNLAB EXPERIMENT**

Run Remorseful Prober against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Detective

**CORE IDEA**

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

**STRATEGIC CHARACTER**

Hunters, Probers, and Opponent Detection

**JINNLAB EXPERIMENT**

Run Detective against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Collective Strategy

### CORE IDEA

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

### STRATEGIC CHARACTER

Hunters, Probers, and Opponent Detection

### JINNLAB EXPERIMENT

Run Collective Strategy against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Handshake

### CORE IDEA

Opponent-modeling strategy that probes for recognizable behavior and changes policy when a pattern is detected.

### STRATEGIC CHARACTER

Hunters, Probers, and Opponent Detection

### JINNLAB EXPERIMENT

Run Handshake against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Memory-One and Outcome-Responsive Strategies

| State | Probability/decision for next cooperation           |
|-------|-----------------------------------------------------|
| CC    | Response after mutual cooperation                   |
| CD    | Response after you cooperated and opponent defected |
| DC    | Response after you defected and opponent cooperated |
| DD    | Response after mutual defection                     |

**Figure 11.1:** Memory-one strategies compress the previous round into four outcome states. Different probability assignments create radically different behavior.

Strategy whose next action is driven primarily by the immediately preceding outcome.

## Memory One Player

**CORE IDEA**

Strategy whose next action is driven primarily by the immediately preceding outcome.

**STRATEGIC CHARACTER**

Memory-One and Outcome-Responsive Strategies

**JINNLAB EXPERIMENT**

Run Memory One Player against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Reactive Player

### CORE IDEA

Strategy whose next action is driven primarily by the immediately preceding outcome.

### STRATEGIC CHARACTER

Memory-One and Outcome-Responsive Strategies

### JINNLAB EXPERIMENT

Run Reactive Player against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## ALLCorALLD

### CORE IDEA

Strategy whose next action is driven primarily by the immediately preceding outcome.

### STRATEGIC CHARACTER

Memory-One and Outcome-Responsive Strategies

### JINNLAB EXPERIMENT

Run ALLCorALLD against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Firm But Fair

### CORE IDEA

Strategy whose next action is driven primarily by the immediately preceding outcome.

### STRATEGIC CHARACTER

Memory-One and Outcome-Responsive Strategies

### JINNLAB EXPERIMENT

Run Firm But Fair against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## GTFT

### CORE IDEA

Strategy whose next action is driven primarily by the immediately preceding outcome.

### STRATEGIC CHARACTER

Memory-One and Outcome-Responsive Strategies

### JINNLAB EXPERIMENT

Run GTFT against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Soft Joss

### CORE IDEA

Strategy whose next action is driven primarily by the immediately preceding outcome.

### STRATEGIC CHARACTER

Memory-One and Outcome-Responsive Strategies

### JINNLAB EXPERIMENT

Run Soft Joss against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Stochastic Cooperator

### CORE IDEA

Strategy whose next action is driven primarily by the immediately preceding outcome.

### STRATEGIC CHARACTER

Memory-One and Outcome-Responsive Strategies

### JINNLAB EXPERIMENT

Run Stochastic Cooperator against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**|** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Stochastic WSLS

### CORE IDEA

Strategy whose next action is driven primarily by the immediately preceding outcome.

### STRATEGIC CHARACTER

Memory-One and Outcome-Responsive Strategies

### JINNLAB EXPERIMENT

Run Stochastic WSLS against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**|** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Win Stay Lose Shift

### CORE IDEA

Repeat an action after a satisfactory outcome and switch after a poor one.

### STRATEGIC CHARACTER

Outcome-based adaptation

### JINNLAB EXPERIMENT

Compare with Tit For Tat under noise; focus on error recovery.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Win Shift Lose Stay

### CORE IDEA

Strategy whose next action is driven primarily by the immediately preceding outcome.

### STRATEGIC CHARACTER

Memory-One and Outcome-Responsive Strategies

### JINNLAB EXPERIMENT

Run Win Shift Lose Stay against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Memory-Two and Longer-Memory Logic

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Strategy that uses a longer short-term history to make punishment, recovery, or coordination more precise.

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### Memory Two Player

#### CORE IDEA

Strategy that uses a longer short-term history to make punishment, recovery, or coordination more precise.

#### STRATEGIC CHARACTER

Memory-Two and Longer-Memory Logic

#### JINNLAB EXPERIMENT

Run Memory Two Player against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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### MEM2

### CORE IDEA

Strategy that uses a longer short-term history to make punishment, recovery, or coordination more precise.

### STRATEGIC CHARACTER

Memory-Two and Longer-Memory Logic

### JINNLAB EXPERIMENT

Run MEM2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## AON2

### CORE IDEA

Strategy that uses a longer short-term history to make punishment, recovery, or coordination more precise.

### STRATEGIC CHARACTER

Memory-Two and Longer-Memory Logic

### JINNLAB EXPERIMENT

Run AON2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Delayed AON1

**CORE IDEA**

Strategy that uses a longer short-term history to make punishment, recovery, or coordination more precise.

**STRATEGIC CHARACTER**

Memory-Two and Longer-Memory Logic

**JINNLAB EXPERIMENT**

Run Delayed AON1 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Finite-State Machines

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Strategy encoded as a state machine: history moves the player between states that determine future actions.

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### FSM Player

#### CORE IDEA

Strategy encoded as a state machine: history moves the player between states that determine future actions.

#### STRATEGIC CHARACTER

Finite-State Machines

#### JINNLAB EXPERIMENT

Run FSM Player against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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### Evolvable FSM Player

#### CORE IDEA

Strategy encoded as a state machine: history moves the player between states that determine future actions.

**STRATEGIC CHARACTER**

Finite-State Machines

**JINNLAB EXPERIMENT**

Run Evolvable FSM Player against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

**Evolved FSM 4****CORE IDEA**

Strategy encoded as a state machine: history moves the player between states that determine future actions.

**STRATEGIC CHARACTER**

Finite-State Machines

**JINNLAB EXPERIMENT**

Run Evolved FSM 4 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

**Evolved FSM 6****CORE IDEA**

Strategy encoded as a state machine: history moves the player between states that determine future actions.

## STRATEGIC CHARACTER

Finite-State Machines

## JINNLAB EXPERIMENT

Run Evolved FSM 6 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Evolved FSM 16

### CORE IDEA

Strategy encoded as a state machine: history moves the player between states that determine future actions.

### STRATEGIC CHARACTER

Finite-State Machines

### JINNLAB EXPERIMENT

Run Evolved FSM 16 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Evolved FSM 16 Noise 05

### CORE IDEA

Strategy encoded as a state machine: history moves the player between states that determine future actions.

**STRATEGIC CHARACTER**

Finite-State Machines

**JINNLAB EXPERIMENT**

Run Evolved FSM 16 Noise 05 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Fortress 3**
**CORE IDEA**

Strategy encoded as a state machine: history moves the player between states that determine future actions.

**STRATEGIC CHARACTER**

Finite-State Machines

**JINNLAB EXPERIMENT**

Run Fortress 3 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Fortress 4**
**CORE IDEA**

Strategy encoded as a state machine: history moves the player between states that determine future actions.

## STRATEGIC CHARACTER

Finite-State Machines

## JINNLAB EXPERIMENT

Run Fortress 4 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Predator

### CORE IDEA

Strategy encoded as a state machine: history moves the player between states that determine future actions.

## STRATEGIC CHARACTER

Finite-State Machines

## JINNLAB EXPERIMENT

Run Predator against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Pun1

### CORE IDEA

Strategy encoded as a state machine: history moves the player between states that determine future actions.



**STRATEGIC CHARACTER**

Finite-State Machines

**JINNLAB EXPERIMENT**

Run Pun1 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Raider

**CORE IDEA**

Strategy encoded as a state machine: history moves the player between states that determine future actions.

**STRATEGIC CHARACTER**

Finite-State Machines

**JINNLAB EXPERIMENT**

Run Raider against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Ripoff

**CORE IDEA**

Strategy encoded as a state machine: history moves the player between states that determine future actions.

## STRATEGIC CHARACTER

Finite-State Machines

## JINNLAB EXPERIMENT

Run Ripoff against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Solution B1

### CORE IDEA

Strategy encoded as a state machine: history moves the player between states that determine future actions.

## STRATEGIC CHARACTER

Finite-State Machines

## JINNLAB EXPERIMENT

Run Solution B1 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Solution B5

### CORE IDEA

Strategy encoded as a state machine: history moves the player between states that determine future actions.

**STRATEGIC CHARACTER**

Finite-State Machines

**JINNLAB EXPERIMENT**

Run Solution B5 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

**TF1****CORE IDEA**

Strategy encoded as a state machine: history moves the player between states that determine future actions.

**STRATEGIC CHARACTER**

Finite-State Machines

**JINNLAB EXPERIMENT**

Run TF1 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

**TF2****CORE IDEA**

Strategy encoded as a state machine: history moves the player between states that determine future actions.

## STRATEGIC CHARACTER

Finite-State Machines

## JINNLAB EXPERIMENT

Run TF2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## TF3

### CORE IDEA

Strategy encoded as a state machine: history moves the player between states that determine future actions.

## STRATEGIC CHARACTER

Finite-State Machines

## JINNLAB EXPERIMENT

Run TF3 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Thumper

### CORE IDEA

Strategy encoded as a state machine: history moves the player between states that determine future actions.

**STRATEGIC CHARACTER**

Finite-State Machines

**JINNLAB EXPERIMENT**

Run Thumper against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Usually Cooperates

**CORE IDEA**

Strategy encoded as a state machine: history moves the player between states that determine future actions.

**STRATEGIC CHARACTER**

Finite-State Machines

**JINNLAB EXPERIMENT**

Run Usually Cooperates against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Usually Defects

**CORE IDEA**

Strategy encoded as a state machine: history moves the player between states that determine future actions.

## STRATEGIC CHARACTER

### Finite-State Machines

## JINNLAB EXPERIMENT

Run Usually Defects against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Lookup Tables, Gamblers, and Hidden-State Models

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Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

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### LookerUp

#### CORE IDEA

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

#### STRATEGIC CHARACTER

Lookup Tables, Gamblers, and Hidden-State Models

#### JINNLAB EXPERIMENT

Run LookerUp against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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### Evolvable LookerUp

**CORE IDEA**

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

**STRATEGIC CHARACTER**

Lookup Tables, Gamblers, and Hidden-State Models

**JINNLAB EXPERIMENT**

Run Evolvable LookerUp against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Evolved LookerUp 1 1 1**

**CORE IDEA**

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

**STRATEGIC CHARACTER**

Lookup Tables, Gamblers, and Hidden-State Models

**JINNLAB EXPERIMENT**

Run Evolved LookerUp 1 1 1 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Evolved LookerUp 2 2 2**



**CORE IDEA**

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

**STRATEGIC CHARACTER**

Lookup Tables, Gamblers, and Hidden-State Models

**JINNLAB EXPERIMENT**

Run Evolved LookerUp 2 2 2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Winner 12****CORE IDEA**

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

**STRATEGIC CHARACTER**

Lookup Tables, Gamblers, and Hidden-State Models

**JINNLAB EXPERIMENT**

Run Winner 12 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Winner 21**

## CORE IDEA

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

## STRATEGIC CHARACTER

Lookup Tables, Gamblers, and Hidden-State Models

## JINNLAB EXPERIMENT

Run Winner 21 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Gambler

## CORE IDEA

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

## STRATEGIC CHARACTER

Lookup Tables, Gamblers, and Hidden-State Models

## JINNLAB EXPERIMENT

Run Gambler against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Evolvable Gambler

**CORE IDEA**

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

**STRATEGIC CHARACTER**

Lookup Tables, Gamblers, and Hidden-State Models

**JINNLAB EXPERIMENT**

Run Evolvable Gambler against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**PSO Gambler 1 1 1****CORE IDEA**

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

**STRATEGIC CHARACTER**

Lookup Tables, Gamblers, and Hidden-State Models

**JINNLAB EXPERIMENT**

Run PSO Gambler 1 1 1 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**PSO Gambler 2 2 2**

### CORE IDEA

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

### STRATEGIC CHARACTER

Lookup Tables, Gamblers, and Hidden-State Models

### JINNLAB EXPERIMENT

Run PSO Gambler 2 2 2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## PSO Gambler 2 2 2 Noise 05

### CORE IDEA

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

### STRATEGIC CHARACTER

Lookup Tables, Gamblers, and Hidden-State Models

### JINNLAB EXPERIMENT

Run PSO Gambler 2 2 2 Noise 05 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## PSO Gambler Mem1

**CORE IDEA**

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

**STRATEGIC CHARACTER**

Lookup Tables, Gamblers, and Hidden-State Models

**JINNLAB EXPERIMENT**

Run PSO Gambler Mem1 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## HMM Player

**CORE IDEA**

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

**STRATEGIC CHARACTER**

Lookup Tables, Gamblers, and Hidden-State Models

**JINNLAB EXPERIMENT**

Run HMM Player against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Evolvable HMM Player

### CORE IDEA

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

### STRATEGIC CHARACTER

Lookup Tables, Gamblers, and Hidden-State Models

### JINNLAB EXPERIMENT

Run Evolvable HMM Player against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Evolved HMM 5

### CORE IDEA

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

### STRATEGIC CHARACTER

Lookup Tables, Gamblers, and Hidden-State Models

### JINNLAB EXPERIMENT

Run Evolved HMM 5 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## ZDMem2

**CORE IDEA**

Parameterized strategy that maps remembered interaction patterns to actions, often with stochastic or evolved variants.

**STRATEGIC CHARACTER**

Lookup Tables, Gamblers, and Hidden-State Models

**JINNLAB EXPERIMENT**

Run ZDMem2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Learning and Adaptation

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Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

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### Adaptive

#### CORE IDEA

Begins with a prescribed sequence and repeatedly favors moves that have produced better outcomes.

#### STRATEGIC CHARACTER

Adaptive reinforcement

#### JINNLAB EXPERIMENT

Compare behavior at 20, 200, and 2,000 turns to see how time horizon changes learning.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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### Adaptor

#### CORE IDEA

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.



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**STRATEGIC CHARACTER**

Learning and Adaptation

**JINNLAB EXPERIMENT**

Run Adaptor against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Adaptor Brief****CORE IDEA**

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

**STRATEGIC CHARACTER**

Learning and Adaptation

**JINNLAB EXPERIMENT**

Run Adaptor Brief against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Adaptor Long****CORE IDEA**

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

## STRATEGIC CHARACTER

Learning and Adaptation

## JINNLAB EXPERIMENT

Run Adaptor Long against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Adaptive Pavlov 2006

### CORE IDEA

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

### STRATEGIC CHARACTER

Learning and Adaptation

### JINNLAB EXPERIMENT

Run Adaptive Pavlov 2006 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Adaptive Pavlov 2011

### CORE IDEA

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

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**STRATEGIC CHARACTER**

Learning and Adaptation

**JINNLAB EXPERIMENT**

Run Adaptive Pavlov 2011 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Bush Mosteller****CORE IDEA**

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

**STRATEGIC CHARACTER**

Learning and Adaptation

**JINNLAB EXPERIMENT**

Run Bush Mosteller against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**DBS****CORE IDEA**

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

## STRATEGIC CHARACTER

Learning and Adaptation

## JINNLAB EXPERIMENT

Run DBS against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Revised Downing

### CORE IDEA

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

### STRATEGIC CHARACTER

Learning and Adaptation

### JINNLAB EXPERIMENT

Run Revised Downing against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Risky QLearner

### CORE IDEA

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

**STRATEGIC CHARACTER**

Learning and Adaptation

**JINNLAB EXPERIMENT**

Run Risky QLearner against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Arrogant QLearner

**CORE IDEA**

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

**STRATEGIC CHARACTER**

Learning and Adaptation

**JINNLAB EXPERIMENT**

Run Arrogant QLearner against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Cautious QLearner

**CORE IDEA**

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

## STRATEGIC CHARACTER

Learning and Adaptation

## JINNLAB EXPERIMENT

Run Cautious QLearner against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Hesitant QLearner

### CORE IDEA

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

## STRATEGIC CHARACTER

Learning and Adaptation

## JINNLAB EXPERIMENT

Run Hesitant QLearner against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Darwin

### CORE IDEA

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

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**STRATEGIC CHARACTER**

Learning and Adaptation

**JINNLAB EXPERIMENT**

Run Darwin against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Calculator****CORE IDEA**

Adaptive strategy that updates behavior in response to observed outcomes or inferred opponent behavior.

**STRATEGIC CHARACTER**

Learning and Adaptation

**JINNLAB EXPERIMENT**

Run Calculator against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Neural and Evolved Strategies

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Learned or evolved strategy whose behavior emerges from optimization rather than a short hand-written rule.

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### Artificial Neural Network based strategy

#### CORE IDEA

Learned or evolved strategy whose behavior emerges from optimization rather than a short hand-written rule.

#### STRATEGIC CHARACTER

Neural and Evolved Strategies

#### JINNLAB EXPERIMENT

Run Artificial Neural Network based strategy against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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### Evolvable ANN



**CORE IDEA**

Learned or evolved strategy whose behavior emerges from optimization rather than a short hand-written rule.

**STRATEGIC CHARACTER**

Neural and Evolved Strategies

**JINNLAB EXPERIMENT**

Run Evolvable ANN against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Evolved ANN

**CORE IDEA**

Learned or evolved strategy whose behavior emerges from optimization rather than a short hand-written rule.

**STRATEGIC CHARACTER**

Neural and Evolved Strategies

**JINNLAB EXPERIMENT**

Run Evolved ANN against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Evolved ANN 5

## CORE IDEA

Learned or evolved strategy whose behavior emerges from optimization rather than a short hand-written rule.

## STRATEGIC CHARACTER

Neural and Evolved Strategies

## JINNLAB EXPERIMENT

Run Evolved ANN 5 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Evolved ANN Noise 5

## CORE IDEA

Learned or evolved strategy whose behavior emerges from optimization rather than a short hand-written rule.

## STRATEGIC CHARACTER

Neural and Evolved Strategies

## JINNLAB EXPERIMENT

Run Evolved ANN Noise 5 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# EvolvedAttention

**CORE IDEA**

Uses a learned attention mechanism over game history.

**STRATEGIC CHARACTER**

Machine-learned policy

**JINNLAB EXPERIMENT**

Compare it with simple strategies across a diverse tournament field; do not infer its internal rule from one match.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Meta-Strategies and Ensembles

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Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

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### Meta Player

#### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

#### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

#### JINNLAB EXPERIMENT

Run Meta Player against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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### Meta Mixer

**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run Meta Mixer against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Hunter

**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run Meta Hunter against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Hunter Aggressive

### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

### JINNLAB EXPERIMENT

Run Meta Hunter Aggressive against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Majority

### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

### JINNLAB EXPERIMENT

Run Meta Majority against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Majority Finite Memory

**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run Meta Majority Finite Memory against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Majority Long Memory

**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run Meta Majority Long Memory against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Majority Memory One

### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

### JINNLAB EXPERIMENT

Run Meta Majority Memory One against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Minority

### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

### JINNLAB EXPERIMENT

Run Meta Minority against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Winner



**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run Meta Winner against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Winner Deterministic

**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run Meta Winner Deterministic against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Winner Ensemble

### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

### JINNLAB EXPERIMENT

Run Meta Winner Ensemble against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Winner Finite Memory

### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

### JINNLAB EXPERIMENT

Run Meta Winner Finite Memory against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Winner Long Memory

**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run Meta Winner Long Memory against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Winner Memory One

**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run Meta Winner Memory One against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Meta Winner Stochastic

### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

### JINNLAB EXPERIMENT

Run Meta Winner Stochastic against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Nice Meta Winner

### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

### JINNLAB EXPERIMENT

Run Nice Meta Winner against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Nice Meta Winner Ensemble

**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run Nice Meta Winner Ensemble against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## NMW E Deterministic

**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run NMW E Deterministic against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## NMWE Finite Memory

### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

### JINNLAB EXPERIMENT

Run NMWE Finite Memory against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## NMWE Long Memory

### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

### JINNLAB EXPERIMENT

Run NMWE Long Memory against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## NMWE Memory One

**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run NMWE Memory One against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## NMWE Stochastic

**CORE IDEA**

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

**STRATEGIC CHARACTER**

Meta-Strategies and Ensembles

**JINNLAB EXPERIMENT**

Run NMWE Stochastic against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Memory Decay

### CORE IDEA

Ensemble strategy that delegates, aggregates, or compares the behavior of other strategies.

### STRATEGIC CHARACTER

Meta-Strategies and Ensembles

### JINNLAB EXPERIMENT

Run Memory Decay against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Zero-Determinant and Payoff-Control Strategies

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Memory-one payoff-control strategy whose importance lies in constraining long-run payoff relationships.

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### LR Player

#### CORE IDEA

Memory-one payoff-control strategy whose importance lies in constraining long-run payoff relationships.

#### STRATEGIC CHARACTER

Zero-Determinant and Payoff-Control Strategies

#### JINNLAB EXPERIMENT

Run LR Player against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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### ZD Extort 2

### CORE IDEA

Memory-one payoff-control strategy whose importance lies in constraining long-run payoff relationships.

### STRATEGIC CHARACTER

Zero-Determinant and Payoff-Control Strategies

### JINNLAB EXPERIMENT

Run ZD Extort 2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## ZD Extort 2 v2

### CORE IDEA

Memory-one payoff-control strategy whose importance lies in constraining long-run payoff relationships.

### STRATEGIC CHARACTER

Zero-Determinant and Payoff-Control Strategies

### JINNLAB EXPERIMENT

Run ZD Extort 2 v2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## ZD Extort 3

**CORE IDEA**

Memory-one payoff-control strategy whose importance lies in constraining long-run payoff relationships.

**STRATEGIC CHARACTER**

Zero-Determinant and Payoff-Control Strategies

**JINNLAB EXPERIMENT**

Run ZD Extort 3 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## ZD Extort 4

**CORE IDEA**

Memory-one payoff-control strategy whose importance lies in constraining long-run payoff relationships.

**STRATEGIC CHARACTER**

Zero-Determinant and Payoff-Control Strategies

**JINNLAB EXPERIMENT**

Run ZD Extort 4 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## ZD Extortion

### CORE IDEA

A zero-determinant strategy designed to impose a payoff relationship favorable to itself.

### STRATEGIC CHARACTER

Payoff control

### JINNLAB EXPERIMENT

Compare pairwise payoff relationships and then place it in a population tournament to distinguish pairwise leverage from ecological success.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## ZD Gen 2

### CORE IDEA

Memory-one payoff-control strategy whose importance lies in constraining long-run payoff relationships.

### STRATEGIC CHARACTER

Zero-Determinant and Payoff-Control Strategies

### JINNLAB EXPERIMENT

Run ZD Gen 2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## ZD GTFT 2

### CORE IDEA

Memory-one payoff-control strategy whose importance lies in constraining long-run payoff relationships.

### STRATEGIC CHARACTER

Zero-Determinant and Payoff-Control Strategies

### JINNLAB EXPERIMENT

Run ZD GTFT 2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## ZD Mischief

### CORE IDEA

Memory-one payoff-control strategy whose importance lies in constraining long-run payoff relationships.

### STRATEGIC CHARACTER

Zero-Determinant and Payoff-Control Strategies

### JINNLAB EXPERIMENT

Run ZD Mischief against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## ZD Set 2

### CORE IDEA

Memory-one payoff-control strategy whose importance lies in constraining long-run payoff relationships.

### STRATEGIC CHARACTER

Zero-Determinant and Payoff-Control Strategies

### JINNLAB EXPERIMENT

Run ZD Set 2 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Axelrod First Tournament Strategies

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Historical tournament entry reconstructed from Axelrod's first competition descriptions.

The first tournament module is based on the descriptions in Axelrod's 1980 Journal of Conflict Resolution paper; the library documentation notes that some original descriptions are incomplete and therefore require explicit implementation assumptions.

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### Anonymous

#### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

#### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

#### JINNLAB EXPERIMENT

Run Anonymous against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Davis

### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

### JINNLAB EXPERIMENT

Run Davis against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Downing

### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

### JINNLAB EXPERIMENT

Run Downing against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Feld

### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

### JINNLAB EXPERIMENT

Run Feld against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Graaskamp

### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

### JINNLAB EXPERIMENT

Run Graaskamp against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Grofman

### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

### JINNLAB EXPERIMENT

Run Grofman against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Joss

### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

### JINNLAB EXPERIMENT

Run Joss against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Nydegger

### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

### JINNLAB EXPERIMENT

Run Nydegger against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Shubik

### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

### JINNLAB EXPERIMENT

Run Shubik against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Stein and Rapoport

### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

### JINNLAB EXPERIMENT

Run Stein and Rapoport against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Tideman and Chieruzzi

### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

### JINNLAB EXPERIMENT

Run Tideman and Chieruzzi against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Tullock

### CORE IDEA

Historical tournament entry reconstructed from Axelrod's first competition descriptions.

### STRATEGIC CHARACTER

Axelrod First Tournament Strategies

### JINNLAB EXPERIMENT

Run Tullock against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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# Axelrod Second Tournament Strategies

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Historical tournament entry from Axelrod's second competition family.

These entries should be read historically as well as strategically: they represent a diverse field of hand-designed responses to repeated cooperation and defection.

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### Appold

#### CORE IDEA

Historical tournament entry from Axelrod's second competition family.

#### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

#### JINNLAB EXPERIMENT

Run Appold against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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### Black

**CORE IDEA**

Historical tournament entry from Axelrod's second competition family.

**STRATEGIC CHARACTER**

Axelrod Second Tournament Strategies

**JINNLAB EXPERIMENT**

Run Black against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Borufsen

**CORE IDEA**

Historical tournament entry from Axelrod's second competition family.

**STRATEGIC CHARACTER**

Axelrod Second Tournament Strategies

**JINNLAB EXPERIMENT**

Run Borufsen against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Cave

**CORE IDEA**

Historical tournament entry from Axelrod's second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Cave against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Champion

### CORE IDEA

Historical tournament entry from Axelrod's second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Champion against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

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## Colbert

### CORE IDEA

Historical tournament entry from Axelrod's second competition family.



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**STRATEGIC CHARACTER**

Axelrod Second Tournament Strategies

**JINNLAB EXPERIMENT**

Run Colbert against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Eatherley****CORE IDEA**

Historical tournament entry from Axelrod’s second competition family.

**STRATEGIC CHARACTER**

Axelrod Second Tournament Strategies

**JINNLAB EXPERIMENT**

Run Eatherley against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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**Getzler****CORE IDEA**

Historical tournament entry from Axelrod’s second competition family.

**STRATEGIC CHARACTER**

## Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Getzler against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Gladstein

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Gladstein against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Graaskamp-Katzen

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

**JINNLAB EXPERIMENT**

Run Graaskamp-Katzen against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Grofman (Second Tournament)

**CORE IDEA**

Historical tournament entry from Axelrod’s second competition family.

**STRATEGIC CHARACTER**

Axelrod Second Tournament Strategies

**JINNLAB EXPERIMENT**

Run Grofman (Second Tournament) against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Harrington

**CORE IDEA**

Historical tournament entry from Axelrod’s second competition family.

**STRATEGIC CHARACTER**

Axelrod Second Tournament Strategies

**JINNLAB EXPERIMENT**

Run Harrington against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**|** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## **Kluepfel**

### **CORE IDEA**

Historical tournament entry from Axelrod’s second competition family.

### **STRATEGIC CHARACTER**

Axelrod Second Tournament Strategies

### **JINNLAB EXPERIMENT**

Run Kluepfel against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**|** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## **Leyvraz**

### **CORE IDEA**

Historical tournament entry from Axelrod’s second competition family.

### **STRATEGIC CHARACTER**

Axelrod Second Tournament Strategies

### **JINNLAB EXPERIMENT**

Run Leyvraz against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see

whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

---

## Mikkelson

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Mikkelson against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

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## Richard Hufford

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Richard Hufford against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Rowsam

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Rowsam against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Tester

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Tester against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

---

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Tideman and Chieruzzi (Second Tournament)

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Tideman and Chieruzzi (Second Tournament) against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Tranquilizer

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Tranquilizer against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Weiner

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Weiner against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## White

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run White against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.



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*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Wm Adams

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Wm Adams against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

## Yamachi

### CORE IDEA

Historical tournament entry from Axelrod’s second competition family.

### STRATEGIC CHARACTER

Axelrod Second Tournament Strategies

### JINNLAB EXPERIMENT

Run Yamachi against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

I
*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

# Specialized and Experimental Strategies

---

Specialized strategy that explores a distinct behavioral mechanism within repeated Prisoner's Dilemma play.

---

### Back Stabber

#### CORE IDEA

Specialized strategy that explores a distinct behavioral mechanism within repeated Prisoner's Dilemma play.

#### STRATEGIC CHARACTER

Specialized and Experimental Strategies

#### JINNLAB EXPERIMENT

Run Back Stabber against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

---

### Double Crosser

### CORE IDEA

Specialized strategy that explores a distinct behavioral mechanism within repeated Prisoner's Dilemma play.

### STRATEGIC CHARACTER

Specialized and Experimental Strategies

### JINNLAB EXPERIMENT

Run Double Crosser against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

---

## Resurrection

### CORE IDEA

Specialized strategy that explores a distinct behavioral mechanism within repeated Prisoner's Dilemma play.

### STRATEGIC CHARACTER

Specialized and Experimental Strategies

### JINNLAB EXPERIMENT

Run Resurrection against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

---

## Double Resurrection

**CORE IDEA**

Specialized strategy that explores a distinct behavioral mechanism within repeated Prisoner's Dilemma play.

**STRATEGIC CHARACTER**

Specialized and Experimental Strategies

**JINNLAB EXPERIMENT**

Run Double Resurrection against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

---

**Fortress 3**
**CORE IDEA**

Specialized strategy that explores a distinct behavioral mechanism within repeated Prisoner's Dilemma play.

**STRATEGIC CHARACTER**

Specialized and Experimental Strategies

**JINNLAB EXPERIMENT**

Run Fortress 3 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

*Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy "best" unless the claim states the exact environment.*

---

**Fortress 4**

### CORE IDEA

Specialized strategy that explores a distinct behavioral mechanism within repeated Prisoner's Dilemma play.

### STRATEGIC CHARACTER

Specialized and Experimental Strategies

### JINNLAB EXPERIMENT

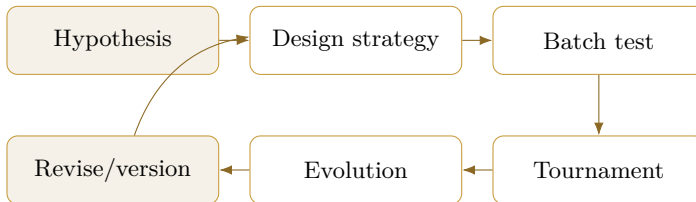
Run Fortress 4 against Tit For Tat, Defector, and Cooperator with at least 10 repetitions and several seeds. Then place it in a small tournament to see whether pairwise behavior generalizes to a mixed field.

**I** *Interpretation rule: compare payoff together with cooperation rate, repeat across seeds when stochastic, and avoid calling the strategy “best” unless the claim states the exact environment.*

---

# Designing Richmack Strategies in JinnLab

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**Figure 22.1:** The Richmack design loop: make a claim, encode a strategy, test it across contexts, then revise with versioned evidence.

JinnLab’s Strategy Designer turns several recurring ideas in this book into explicit parameters: opening move, retaliation length, forgiveness threshold, and random defection probability. This is deliberately simpler than the full Axelrod strategy interface, but it is ideal for hypothesis-driven experimentation.

## Opening move

---

Choose cooperation when you want to offer trust immediately; choose defection when you want suspicion, probing, or protection against early exploitation.

## Retaliation length

---

Longer retaliation increases deterrence but raises the risk of locking into conflict after mistakes.

## Forgiveness threshold

---

Forgiveness is an error-correction mechanism. It can recover cooperation, but excessive forgiveness can invite repeated exploitation.

## Random defection

---

A small stochastic component can probe opponents or break deterministic cycles, but it can also destroy stable cooperation.

Recommended workflow: save a family name, create versioned variants, run pairwise matches, run a batch across seeds, then place the variants in the same tournament. Keep a hypothesis and conclusion in the JinnLab notebook.



# The Richmack Experimental Notebook

---

|                                   |
|-----------------------------------|
| Experiment ID .....               |
| Hypothesis .....                  |
| Strategies / versions .....       |
| Turns / repetitions / seeds ..... |
| Observed result .....             |
| Interpretation .....              |
| Next test .....                   |

**Figure 23.1:** A compact notebook template for turning simulations into reproducible evidence.

## Experiment Template

---

Question: .....

Hypothesis: .....

Strategies: .....

Turns: .....

Repetitions: .....

Seed range: .....

Noise / mutation assumptions: .....  
 Observed payoff pattern: .....  
 Observed cooperation pattern: .....  
 Tournament / evolution result: .....  
 Conclusion: .....  
 What would falsify this conclusion? .....

## Interpretation Checklist

---

- ☐ Did I compare more than one opponent?
- ☐ Did I use multiple seeds for stochastic strategies?
- ☐ Is the score difference large relative to run-to-run variation?
- ☐ Does the result survive a change in match length?
- ☐ Does the result survive noise?
- ☐ Does tournament success agree with evolutionary survival?
- ☐ Am I describing the environment in the conclusion?

# Strategy Catalog Used in This Edition

---

The entries above are organized for learning rather than alphabetically. This appendix provides a quick catalog of the named strategies covered in the book. Axelrod 4.14.0 reports 243 strategies in its default strategy collection; some library entries are parameterized or specialized variants whose public display names differ from Python class names.

## Reciprocity: Tit-for-Tat and Its Descendants

---

- Tit For Tat
- Tit For 2 Tats
- Two Tits For Tat
- Hard Tit For Tat
- Hard Tit For 2 Tats
- Suspicious Tit For Tat
- Anti Tit For Tat
- Bully
- Contrite Tit For Tat
- Adaptive Tit For Tat
- Random Tit For Tat
- Sneaky Tit For Tat

- Spiteful Tit For Tat
- Dynamic Two Tits For Tat
- Slow Tit For Two Tats 2
- Omega TFT
- Gradual
- Original Gradual
- Alexei
- Michaelos
- Eugene Nier
- Burn Both Ends
- N Tits For M Tats

## Cooperation, Defection, and Simple Baselines

---

- Cooperator
- Defector
- Random
- Alternator
- Tricky Cooperator
- Tricky Defector
- Better and Better
- Very Bad
- Short Mem
- Inverse
- Negation
- Willing
- Hopeless
- Desperate
- Appeaser

- Doubler
- Momentum

## Grudges, Punishment, and Retaliation

---

- Grudger
- Forgetful Grudger
- Soft Grudger
- General Soft Grudger
- Grudger Alternator
- Opposite Grudger
- Aggravater
- CAPRI
- Easy Go
- Spiteful CC
- Gradual Killer
- Punisher
- Inverse Punisher
- Level Punisher
- Tricky Level Punisher
- Retaliate
- Retaliate 2
- Retaliate 3
- Limited Retaliate
- Limited Retaliate 2
- Limited Retaliate 3
- Fool Me Once
- Forgetful Fool Me Once
- Once Bitten

## Majority, Frequency, and Population Reading

---

- Go By Majority
- Go By Majority 5
- Go By Majority 10
- Go By Majority 20
- Go By Majority 40
- Hard Go By Majority
- Hard Go By Majority 5
- Hard Go By Majority 10
- Hard Go By Majority 20
- Hard Go By Majority 40
- Frequency Analyzer
- Average Copier
- Nice Average Copier
- SelfSteem
- Grumpy
- Stalker

## Patterns, Cycles, and Mathematical Sequences

---

- Cyclor
- Cyclor DC
- Cyclor DDC
- Cyclor CCD
- Cyclor CCCD
- Cyclor CCCDCD
- Cyclor CCCCCD

- AntiCycler
- Evolvable Cycler
- Sequence Player
- Thue-Morse
- Thue-Morse Inverse
- Co to Do Ratio
- Golden
- Pi
- e

## Hunters, Probers, and Opponent Detection

---

- Alternator Hunter
- Cooperator Hunter
- Cycle Hunter
- Defector Hunter
- Eventual Cycle Hunter
- Math Constant Hunter
- Random Hunter
- Prober
- Prober 2
- Prober 3
- Prober 4
- Hard Prober
- Naive Prober
- Remorseful Prober
- Detective
- Collective Strategy
- Handshake

## Memory-One and Outcome-Responsive Strategies

---

- Memory One Player
- Reactive Player
- ALLCorALLD
- Firm But Fair
- GTFT
- Soft Joss
- Stochastic Cooperator
- Stochastic WSLs
- Win Stay Lose Shift
- Win Shift Lose Stay

## Memory-Two and Longer-Memory Logic

---

- Memory Two Player
- MEM2
- AON2
- Delayed AON1

## Finite-State Machines

---

- FSM Player
- Evolvable FSM Player
- Evolved FSM 4
- Evolved FSM 6
- Evolved FSM 16



- Evolved FSM 16 Noise 05
- Fortress 3
- Fortress 4
- Predator
- Pun1
- Raider
- Ripoff
- Solution B1
- Solution B5
- TF1
- TF2
- TF3
- Thumper
- Usually Cooperates
- Usually Defects

## Lookup Tables, Gamblers, and Hidden-State Models

---

- LookerUp
- Evolvable LookerUp
- Evolved LookerUp 1 1 1
- Evolved LookerUp 2 2 2
- Winner 12
- Winner 21
- Gambler
- Evolvable Gambler
- PSO Gambler 1 1 1
- PSO Gambler 2 2 2
- PSO Gambler 2 2 2 Noise 05

- PSO Gambler Mem1
- HMM Player
- Evolvable HMM Player
- Evolved HMM 5
- ZDMem2

## Learning and Adaptation

---

- Adaptive
- Adaptor
- Adaptor Brief
- Adaptor Long
- Adaptive Pavlov 2006
- Adaptive Pavlov 2011
- Bush Mosteller
- DBS
- Revised Downing
- Risky QLearner
- Arrogant QLearner
- Cautious QLearner
- Hesitant QLearner
- Darwin
- Calculator

## Neural and Evolved Strategies

---

- Artificial Neural Network based strategy
- Evolvable ANN

- Evolved ANN
- Evolved ANN 5
- Evolved ANN Noise 5
- EvolvedAttention

## Meta-Strategies and Ensembles

---

- Meta Player
- Meta Mixer
- Meta Hunter
- Meta Hunter Aggressive
- Meta Majority
- Meta Majority Finite Memory
- Meta Majority Long Memory
- Meta Majority Memory One
- Meta Minority
- Meta Winner
- Meta Winner Deterministic
- Meta Winner Ensemble
- Meta Winner Finite Memory
- Meta Winner Long Memory
- Meta Winner Memory One
- Meta Winner Stochastic
- Nice Meta Winner
- Nice Meta Winner Ensemble
- NMW E Deterministic
- NMWE Finite Memory
- NMWE Long Memory
- NMWE Memory One

- NMWE Stochastic
- Memory Decay

## Zero-Determinant and Payoff-Control Strategies

---

- LR Player
- ZD Extort 2
- ZD Extort 2 v2
- ZD Extort 3
- ZD Extort 4
- ZD Extortion
- ZD Gen 2
- ZD GTFT 2
- ZD Mischief
- ZD Set 2

## Axelrod First Tournament Strategies

---

- Anonymous
- Davis
- Downing
- Feld
- Graaskamp
- Grofman
- Joss
- Nydegger
- Shubik
- Stein and Rapoport

- Tideman and Chieruzzi
- Tullock

## Axelrod Second Tournament Strategies

---

- Appold
- Black
- Borufsen
- Cave
- Champion
- Colbert
- Eatherley
- Getzler
- Gladstein
- Graaskamp-Katzen
- Grofman (Second Tournament)
- Harrington
- Kluepfel
- Leyvraz
- Mikkelsen
- Richard Hufford
- Rowsam
- Tester
- Tideman and Chieruzzi (Second Tournament)
- Tranquilizer
- Weiner
- White
- Wm Adams
- Yamachi

## Specialized and Experimental Strategies

---

- Back Stabber
- Double Crosser
- Resurrection
- Double Resurrection
- Fortress 3
- Fortress 4
- Predator
- Raider
- Ripoff
- Solution B1
- Solution B5
- Usually Cooperates
- Usually Defects

# JinnLab Command Examples

---

## Pairwise match

---

```
jinnlab match "Tit For Tat" Defector -turns 200 -repetitions 20 -seed 42
```

## Tournament

---

```
jinnlab tournament "Tit For Tat" Defector Grudger Cooperator -output csv
```

## Evolution

---

```
jinnlab evolve "Tit For Tat=40" "Defector=30" "Cooperator=30" -generations 100
```

---

# Notes on Sources and Attribution

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Primary reference points for this edition include the Axelrod 4.14.0 strategy index and documentation, the Axelrod project's reproducibility paper, and the historical tournament literature. Strategy descriptions in this book are intentionally paraphrased and pedagogical. For exact implementation details, consult the source code and strategy docstrings for the installed Axelrod version.



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# Selected Bibliography

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# Bibliography

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- [5] Knight, Vincent, Owen Campbell, Marc Harper, T. J. Gaffney, and Nikoleta E. Glynatsi. “Reviving, Reproducing, and Revisiting Axelrod’s Second Tournament.” 2025.

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# Colophon

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This pocket edition was typeset in LaTeX on A5 stock with a restrained black-and-gold visual system. Strategy profiles are framed for rapid reference; figures emphasize conceptual structure rather than invented performance data.